

Anne Perera

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EDUCATION

Universiy of Kelaniya <i>BSc (Hons) in Computer Science</i>	Sri Lanka 2023 - Present
Universiy of Colombo School of Computing <i>Certificate Course on Data Science with Python</i>	Sri Lanka 2023
Visakha Vidyalaya <i>G.C.E. Advanced Level(2021) - Physical Science Stream</i>	Colombo 05, Sri Lanka 2012 - 2020

PROJECTS

SLR GAT(GNN) <i>Python, PyTorch, PyTorch Geometric, GATv2, GNNExplainer, Open-Meteo</i> Ongoing Research - Developing a GATv2 framework with climate-conditioned edge weights to model and explain how monsoon weather reshapes train-delay propagation across a railway network. - Building a spatio-temporal event graph from timetable data and applying GNNExplainer to generate a season-aware cascade risk map for operational decision-support.	
Super-Resolution and Inference Optimization <i>Python, PyTorch, ONNX, TensorRT</i> - Built an image super-resolution pipeline using Real-ESRGAN (RRDBNet) to upscale low-resolution images 4×, with tiled inference for memory-safe processing of large images - Optimized model deployment through a PyTorch → ONNX → TensorRT (FP16) pipeline, reducing inference latency 2× while preserving output quality - Built a three-way benchmark harness measuring p50/p90/p99 latency and PSNR/SSIM/LPIPS across backends, with numerical parity verification at each conversion step	May 2026
CreditAI - Enterprise Risk Intelligence Platform <i>Python, FastAPI, XGBoost, Scikit-learn, Docker</i> - Built an AI-driven credit risk system using ensemble models (XGBoost, Random Forest, Gradient Boosting) with fraud detection and NLP-based sentiment analysis for real-time lending decisions. - Built a modular FastAPI backend with Docker deployment, designed for scalable expansion with FinBERT and LSTM integration.	2026
Citation and Claim Verifier Chrome Extension <i>JavaScript, Manifest V3, Node.js, Vercel, Claude API</i> - Built a Chrome extension that fact-checks highlighted claims, returning source-backed verdicts (Supported / Disputed / Unverifiable) via the Claude API with live web search - Designed a serverless Vercel backend keeping the API key server-side, with token auth and rate limiting to prevent abuse	2026
AgriLinker Web Application <i>Java, Spring Boot, MongoDB, WebSockets, JWT</i> - Built a full-stack farmer-buyer platform with role-based authentication and order management. - Integrated real-time messaging and notifications using WebSockets and SSE. - Developed AI services (chatbot, crop advisor, complaint handling, sentiment analysis).	Feb 2026
Vital — Health and Fitness Companion App <i>Flutter, Dart, Firebase, Cloud Firestore, SQLite</i> - Built a cross-platform mobile app unifying five health domains (step, workout, meal, medication, and sleep tracking) with an offline-first SQLite store and animated Material 3 UI - Engineered a Firebase backend with Google Sign-In authentication and per-user Cloud Firestore sync, layering cross-device cloud backup over local storage while preserving full offline functionality	Dec 2025
Local RAG <i>Python, FastAPI, React, Tailwind CSS, FAISS, SQLite, Ollama, Sentence Transformers</i> - Built a fully local Retrieval-Augmented Generation (RAG) chatbot that answers questions exclusively from user-uploaded PDF, TXT, and Markdown documents using semantic search and grounded responses - Implemented document parsing with PyMuPDF, embedding generation with Sentence Transformers, FAISS-based vector retrieval, SQLite persistence, and a React + Tailwind interface with chat session history and source citations	2026
Medical Appointment Healthcare System <i>C#, .NET Framework, WinForms, SQL Server, RDLC Reports</i>	2025

- Developed a desktop healthcare management system for handling patient records, doctor profiles, appointments, and administrative workflows through an intuitive WinForms interface
- Implemented SQL Server database integration and generated printable RDLC reports to support efficient record management and appointment tracking

ML Based Bank Customer Churn Prediction System | *Pandas, Scikit-learn, Seaborn, Matplotlib* 2023

- Developed a machine learning-based churn prediction system to identify at-risk banking customers and support data-driven retention strategies
- Implemented a Random Forest classification model with full data preprocessing pipeline including feature engineering, label encoding, and feature scaling using StandardScaler

TECHNICAL SKILLS

Languages: Java, Python, C/C++/C#, SQL, PL SQL, TypeScript, JavaScript, HTML/CSS/PHP, Prolog

Frontend Development: React.js, Next.js, React Native, Tailwind CSS, Bootstrap

Backend Development: Spring Boot, Node.js, Express.js, FastAPI, Flask, REST APIs

Database Technologies: MySQL, MongoDB, SQL Server, Oracle, SQLite

AI and Data Science: PyTorch, LangChain, Natural Language Processing (NLP), Sentiment Analysis, Machine Learning

Cloud and DevOps: AWS, Docker, Vercel

CERTIFICATIONS

- Training Course on Data Science with Python
- Python Workshop - PYQUEST '24
- Introduction to Information Security

AWARDS

All Island Rank 6 – Grade 5 Scholarship Examination - Secured 6th place at the All Island level and 3rd place in Gampaha District in the highly competitive National Grade 5 Scholarship Examination

EXTRA CURRICULAR ACTIVITIES

Head of Events and Logistics - AWS Student Community Day - Sri Lanka 2026

Editor - Organizing Committee Member of 14th Installation Ceremony(23/24)

Member - AWS Student Builder Group 25/26, 26/27

Member - Rotaract Club of University of Kelaniya

Organizing Committee Member - Open Day 2025 – Faculty of Science, University of Kelaniya